R16

Code No: 138GY

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, July - 2023
SENSORS AND TRANSDUCERS
(Common to EEE, ECE, CSE)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What is the purpose of measurement?	[2]
b)	List the types of instrumental error.	[3]
c)	What is dynamic error?	[2]
d)	What is repeatability?	[3]
e)	Define the permittivity.	[2]
f)	Write the working principle of proving ring.	[3]
g)	What is the principle of capacitive transducer?	[2]
h)	What are the applications of LVDT?	[3]
i)	Give any applications of smart sensors.	[2]
j)	Define Hall Effect.	[3]

PART – B

(50 Marks)

2.a) What are the three categories of systematic errors in the instrument and explain in detail.
b) Draw the block diagram of measuring system. [5+5]

OR

3.a) Compare the between accuracy and precision.
b) Explain the basic requirements of a sensor. [5+5]

4. Write the equations of various mathematical model of transducer, explain with diagram in detail. [10]

OR

5. Discuss in detail the various static and dynamic characteristics of a measurement system. [10]

6.a) Explain the working of strain gauge.
b) Explain the working of load cell. [5+5]

OR

7. Explain about thermistors materials and its Constructions and Characteristics. [10]

8.a) Compare the self-inductive transducer and mutual inductive transducers.
b) Explain the principle of RVDT with suitable diagram. [5+5]

OR

9.a) Discuss about variable air gap type capacitive transducer with its diagram.
b) Write the working principle of capacitor microphone. [5+5]

10.a) What is MEMS? Explain the advantages, disadvantages and applications of MEMS.
b) Discuss about construction and working of digital transducers. [5+5]

OR

11.a) Explain the construction and working of film sensors with neat sketch.
b) Describe the working of piezoelectric transducer. [5+5]

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